

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

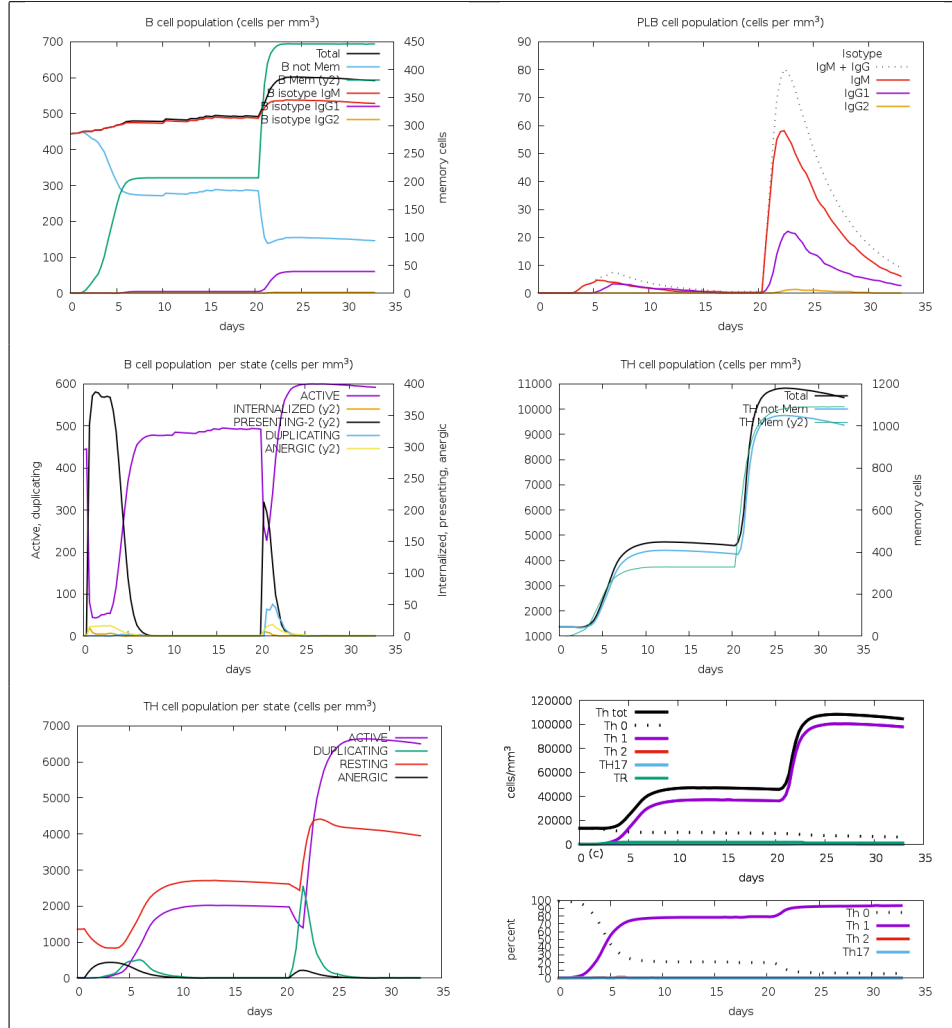


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

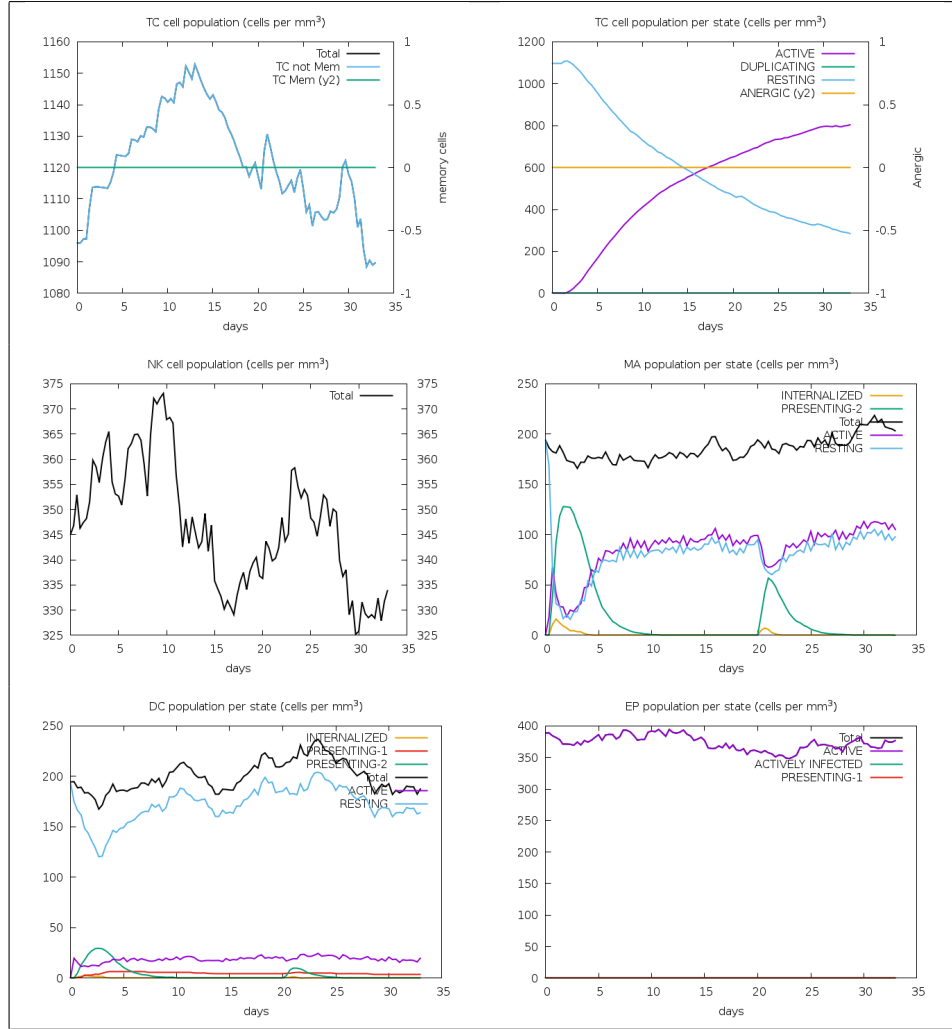


Figure 2: Legend: symbols as figure above.

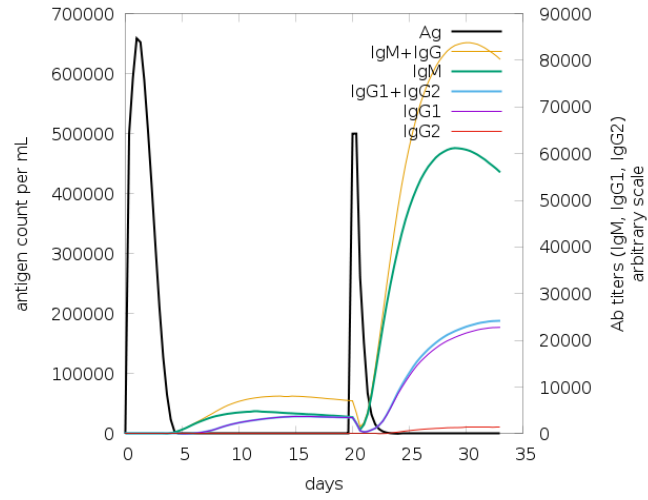


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

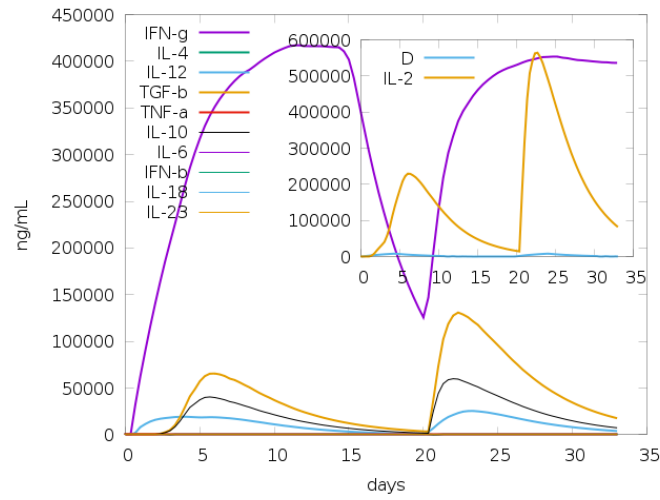


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

AKFVAAWTLKAAAGGSVTVTAFWAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEPPVDIPAVAAAAAIFLIGALFAAAAAVEIPEFPFYAAAAAKLPHWPTPLAAAAAPPVEGAAAEAGDDG
DAAAAADPPFNSILFALLAAAAAANFKLSLSTEFINPLFAAAAASDREVGRGSALYEASTAAAAATSSFVNTNTVIGIEL
PAAAAACPSLVDVSIIPRTPDVLHEDAAAAADGKNKETFHERADSFAAAAAVFSEIPASQDMPNTNTAAAAAGEGNWTSI
DGGTFCF

[illegible]

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1]      pos=58 len=4      AGTD
2]      pos=153 len=9     AAEGDDGDA
3]      pos=205 len=7     ASDREVR
4]      pos=270 len=6     ADGKNK
5]      pos=301 len=4     DMPT

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Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 4 MHC I molecules

Read class I peptide list from file? NO

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

AKFVAAWTLKAAAGGSVTVTAFWAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEPPVDIPAVAAAAAIFLIGALFAAAAAVEIPEFPFYAAAAAKLPHWPTPLAAAAAPPVEGAAAEAGDDG
DAAAAADPPFNSILFALLAAAAAANFKLSLSTEFIPNLFAAAAASDREVGRSALYEASTAAAAATSSFVNTNTVIGIEL
PAAAAACPSLVDVSIIPRTPDVLHEDAAAAADGKNKETFHERADSFAAAAAVFSEIPASQDMPNTNTAAAAAGEGNWTS
DGGTCK

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

5

Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014109_5_WYwskJBWWC5x.FSA_1_001

AKFVAAWTLKAAAGGSVTVTAFWAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEVPDIPAVAAAAAIFLIGFALFAAAAAVEIPEFPFYAAAAAKLPHWPTLAAAAAPPMVEGAAAEGDDG
DAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAASDREVRGSALYEASTAAAAATSSFVTNTTVGIEL
PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFERADSFAAAAAVFSDEIPASQDMPTNTAAAAAGEGNWTS
DGGTFK

Epitopes of protein 0 -----

0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event

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Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014109_5_WYwskJBWWC5x.FSA_1_001

AKFVAAWTLKAAAGGSVTVTAFWAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEVPDIPAVAAAAAIFLIGFALFAAAAAVEIPEFPFYAAAAAKLPHWPTLAAAAAPPMVEGAAAEGDDG
DAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAASDREVRGSALYEASTAAAAATSSFVTNTTVGIEL
PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFERADSFAAAAAVFSDEIPASQDMPTNTAAAAAGEGNWTS
DGGTFK

Epitopes of protein 0 -----

0]	pos= 95	score=0.037638	unnormalised=5.0192000000	IPAVAAAAA
1]	pos= 120	score=0.006151	unnormalised=0.8202000000	IPEFPFYAA
2]	pos= 123	score=0.036836	unnormalised=4.9122000000	FPFYAAAAA
3]	pos= 133	score=0.008183	unnormalised=1.0912000000	LPHWPTLA
4]	pos= 137	score=0.026660	unnormalised=3.5552000000	TPTLAAAAA
5]	pos= 140	score=0.009983	unnormalised=1.3312000000	LAAAAAPPM
6]	pos= 145	score=0.000774	unnormalised=0.1032000000	APPMVEGAA
7]	pos= 196	score=0.018749	unnormalised=2.5002000000	IPNLFAAAA
8]	pos= 221	score=0.013529	unnormalised=1.8042000000	AAAAATSSF
9]	pos= 296	score=0.015269	unnormalised=2.0362000000	IPASQDMPT
10]	pos= 302	score=0.040038	unnormalised=5.3392000000	MPTNTAAAA
11]	pos= -1	score=0.786190	unnormalised=104.8410000000	non-binding event

=====

Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014109_5_WYwskJBWWC5x.FSA_1_001

AKFVAAWTLKAAAGGSVTVTAFWAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEVPDIPAVAAAAAIFLIGFALFAAAAAVEIPEFPFYAAAAAKLPHWPTLAAAAAPPMVEGAAAEGDDG
DAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAASDREVRGSALYEASTAAAAATSSFVTNTTVGIEL

PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFHERADSFAAAAAVFSDEIPASQDMPTNTAAAAAGEGNWTSL
DGGTFK

Epitopes of protein 0 -----

0]	pos=	95	score=0.037638	unnormalised=5.0192000000	IPAVAAAAA
1]	pos=	120	score=0.006151	unnormalised=0.8202000000	IPEFPFYAA
2]	pos=	123	score=0.036836	unnormalised=4.9122000000	FPFYAAAAA
3]	pos=	133	score=0.008183	unnormalised=1.0912000000	LPHWTPTLA
4]	pos=	137	score=0.026660	unnormalised=3.5552000000	TPTLAAAAA
5]	pos=	140	score=0.009983	unnormalised=1.3312000000	LAAAAAPPM
6]	pos=	145	score=0.000774	unnormalised=0.1032000000	APPMVEGAA
7]	pos=	196	score=0.018749	unnormalised=2.5002000000	IPNLFAAAA
8]	pos=	221	score=0.013529	unnormalised=1.8042000000	AAAAATSSF
9]	pos=	296	score=0.015269	unnormalised=2.0362000000	IPASQDMPT
10]	pos=	302	score=0.040038	unnormalised=5.3392000000	MPTNTAAAA
11]	pos=	-1	score=0.786190	unnormalised=104.8410000000	non-binding event

DoPeptideList_II:

Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014109_5_WYwskJBWWC5x.FSA_1_001

AKFVAAWTLKAAAGGSVTVTAFWAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEVPDIPAVAAAAAIFLIGFALFAAAAAVEIPEFPFYAAAAAKLPHWTPTLAAAAAPPMVEGAAAEGDDG
DAAAAADPPFNLSILFALLAAAAAANFKLSSTEFIPNLFAAAAAADREVRGSALYEASTAAAAATSSFVTNTTVGIEL
PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFHERADSFAAAAAVFSDEIPASQDMPTNTAAAAAGEGNWTSL
DGGTFK

Epitopes of protein 0 -----

0]	pos=	2	score=0.010488	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.001430	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	8	score=0.021189	unnormalised=6.8375600000	LKAAAGGSV
3]	pos=	21	score=0.013745	unnormalised=4.4355600000	FWAWAAAAA
4]	pos=	22	score=0.016894	unnormalised=5.4515600000	WAWAAAAAY
5]	pos=	24	score=0.032190	unnormalised=10.3875600000	WAAAAAYTI
6]	pos=	34	score=0.024808	unnormalised=8.0055600000	WTELMAAAA
7]	pos=	35	score=0.002744	unnormalised=0.8855600000	TELMAAAAA
8]	pos=	37	score=0.025763	unnormalised=8.3135600000	LMAAAAANV
9]	pos=	38	score=0.003652	unnormalised=1.1785600000	MAAAAAANV
10]	pos=	51	score=0.018375	unnormalised=5.9295600000	IFRAAAAAG
11]	pos=	52	score=0.039082	unnormalised=12.6115600000	FRAAAAAGT
12]	pos=	64	score=0.012608	unnormalised=4.0685600000	IFLTAAAAA
13]	pos=	65	score=0.023002	unnormalised=7.4225600000	FLTAAAAAA
14]	pos=	80	score=0.024564	unnormalised=7.9265600000	VYRAAAAAA
15]	pos=	81	score=0.047189	unnormalised=15.2275600000	YRAAAAAAL
16]	pos=	95	score=0.006698	unnormalised=2.1615600000	IPAVAAAAA
17]	pos=	98	score=0.014458	unnormalised=4.6655600000	VAAAAAIFL

18]	pos= 105	score=0.008564	unnormalised=2.7635600000	FLIGFALFA
19]	pos= 109	score=0.026751	unnormalised=8.6325600000	FALFAAAAA
20]	pos= 112	score=0.029212	unnormalised=9.4265600000	FAAAAAVEI
21]	pos= 125	score=0.039094	unnormalised=12.6155600000	FYAAAAAKL
22]	pos= 126	score=0.012593	unnormalised=4.0635600000	YAAAAAKLP
23]	pos= 136	score=0.022388	unnormalised=7.2245600000	WPTLAAAA
24]	pos= 140	score=0.027870	unnormalised=8.9935600000	LAAAAAPP
25]	pos= 169	score=0.013302	unnormalised=4.2925600000	FNSILFALL
26]	pos= 173	score=0.031545	unnormalised=10.1795600000	LFALLAAAA
27]	pos= 174	score=0.046581	unnormalised=15.0315600000	FALLAAAA
28]	pos= 176	score=0.029646	unnormalised=9.5665600000	LLAAAAAA
29]	pos= 177	score=0.027907	unnormalised=9.0055600000	LAAAAAAA
30]	pos= 187	score=0.024272	unnormalised=7.8325600000	FKSLSTEF
31]	pos= 196	score=0.000665	unnormalised=0.2145600000	IPNLFAAAA
32]	pos= 200	score=0.018016	unnormalised=5.8135600000	FAAAAAADR
33]	pos= 215	score=0.003708	unnormalised=1.1965600000	LYEASTAAA
34]	pos= 216	score=0.029215	unnormalised=9.4275600000	YEASTAAA
35]	pos= 229	score=0.012552	unnormalised=4.0505600000	FVTNTTVGI
36]	pos= 237	score=0.011229	unnormalised=3.6235600000	IELPAAAA
37]	pos= 261	score=0.005338	unnormalised=1.7225600000	VLHEDAAA
38]	pos= 262	score=0.000336	unnormalised=0.1085600000	LHEDAAAA
39]	pos= 285	score=0.035289	unnormalised=11.3875600000	FAAAAAVFS
40]	pos= -1	score=0.205047	unnormalised=66.1680000000	non-binding event

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Allele: DRB1_0101
Pseudo sequence: KAFHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014109_5_WYwskJBWWC5x.FSA_1_001

AKFVAAWTLKAAAGGSVTVTAFWAWAAAAAYTIPWTELMAAAAANVFDLEGIFRAAAAAAGTDNEIFLTAAAAAESLWVG
VYRAAAAAALAEFPVDIPAVAAAAAIFLIGFALFAAAAAVEIPEFPFYAAAAAKLPHWPTLAAAAAPPMEGAAEGDDG
DAAAAADPPFNLSILFALLAAAAAAANFKSLSTEFIPNLFAAAAASDREVRGSALYEASTAAAAATSSFTNTTVGIEL
PAAAAACPSLDVDSIIPRTPDVLHEDAAAAADGKNKETFHERADSFAAAAAVFSDEIPASQDMPNTAAAAAGEGNWTS
DGGTFK

Epitopes of protein 0 -----

0]	pos= 2	score=0.010488	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001430	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 8	score=0.021189	unnormalised=6.8375600000	LKAAAGGSV
3]	pos= 21	score=0.013745	unnormalised=4.4355600000	FWAWAAAA
4]	pos= 22	score=0.016894	unnormalised=5.4515600000	WAWAAAAAY
5]	pos= 24	score=0.032190	unnormalised=10.3875600000	WAAAAAYTI
6]	pos= 34	score=0.024808	unnormalised=8.0055600000	WTELMAAAA
7]	pos= 35	score=0.002744	unnormalised=0.8855600000	TELMAAAA
8]	pos= 37	score=0.025763	unnormalised=8.3135600000	LMAAAAANV
9]	pos= 38	score=0.003652	unnormalised=1.1785600000	MAAAAAANV
10]	pos= 51	score=0.018375	unnormalised=5.9295600000	IFRAAAAAAG
11]	pos= 52	score=0.039082	unnormalised=12.6115600000	FRAAAAAAGT
12]	pos= 64	score=0.012608	unnormalised=4.0685600000	IFLTAAAA
13]	pos= 65	score=0.023002	unnormalised=7.4225600000	FLTKAAAA
14]	pos= 80	score=0.024564	unnormalised=7.9265600000	VYRAAAAA
15]	pos= 81	score=0.047189	unnormalised=15.2275600000	YRAAAAAAL
16]	pos= 95	score=0.006698	unnormalised=2.1615600000	IPAVAAAA
17]	pos= 98	score=0.014458	unnormalised=4.6655600000	VAAAAAIFL
18]	pos= 105	score=0.008564	unnormalised=2.7635600000	FLIGFALFA
19]	pos= 109	score=0.026751	unnormalised=8.6325600000	FALFAAAAA
20]	pos= 112	score=0.029212	unnormalised=9.4265600000	FAAAAAVEI
21]	pos= 125	score=0.039094	unnormalised=12.6155600000	FYAAAAAKL

22]	pos= 126	score=0.012593	unnormalised=4.0635600000	YAAAAAKLP
23]	pos= 136	score=0.022388	unnormalised=7.2245600000	WTPTLAAAA
24]	pos= 140	score=0.027870	unnormalised=8.9935600000	LAAAAAPPM
25]	pos= 169	score=0.013302	unnormalised=4.2925600000	FNSILFALL
26]	pos= 173	score=0.031545	unnormalised=10.1795600000	LFALLAAAA
27]	pos= 174	score=0.046581	unnormalised=15.0315600000	FALLAAAAA
28]	pos= 176	score=0.029646	unnormalised=9.5665600000	LLAAAAAAA
29]	pos= 177	score=0.027907	unnormalised=9.0055600000	LAAAAAAA
30]	pos= 187	score=0.024272	unnormalised=7.8325600000	FKSLSSTEF
31]	pos= 196	score=0.000665	unnormalised=0.2145600000	IPNLFAAAA
32]	pos= 200	score=0.018016	unnormalised=5.8135600000	FAAAAASDR
33]	pos= 215	score=0.003708	unnormalised=1.1965600000	LYEASTAAA
34]	pos= 216	score=0.029215	unnormalised=9.4275600000	YEASTAAA
35]	pos= 229	score=0.012552	unnormalised=4.0505600000	FVTNTTVGI
36]	pos= 237	score=0.011229	unnormalised=3.6235600000	IELPAAAAA
37]	pos= 261	score=0.005338	unnormalised=1.7225600000	VLHEDAAAA
38]	pos= 262	score=0.000336	unnormalised=0.1085600000	LHEDAAAAA
39]	pos= 285	score=0.035289	unnormalised=11.3875600000	FAAAAAVFS
40]	pos= -1	score=0.205047	unnormalised=66.1680000000	non-binding event
